# Making an Electromagnet

Aim: To build an electromagnet

## Equipment:

- Insulated copper wire
- Wire strippers
- Battery (9V)
- Iron nail (approx. 50mm)
- Crocodile clips - optional
- Paperclips

### Safety Information

Take care when handling and cutting wire to avoid cuts.

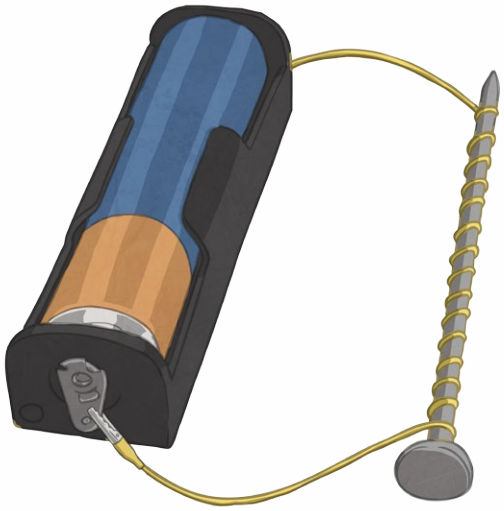
## Method:

Step 1: Wrap the copper wire around the iron nail, leaving about 10cm lengths at each end. Try not to wrap it overly tightly around the nail - you should be able to remove the nail from the coil after wrapping.

Step 2: Carefully strip the insulation from the ends of the wire to expose the copper inside and to make a connection. (Or connect with crocodile clips, if available).

Step 3: Connect the exposed ends, one to each terminal of the battery.

Step 4: Test your electromagnet by observing whether it attracts the paperclips.



### Exploring further

You could investigate how the number of coils in your wire affects the strength of the electromagnet.

You could investigate how the number of batteries connected affects the strength of the electromagnet.

You could investigate how the metal core inside the coil affects the strength of the electromagnet.